Recovery / Fault Injection Test Sheet

Note added in this version: Bug_2 (WDT reset ≈30 s) was withdrawn after mentor feedback. The 30 s was real wall-clock time, because Wokwi slows to ≈2% speed after "hang". By the Wokwi simulation timer, [FAULT] was at ≈11:52.4 and the reboot at ≈11:55.4, i.e. ≈3 s, which meets the requirement.

Webinar 5 — Sleep, WDT, Power Loss, Persistence

Alexandra Holoborodko

Environment

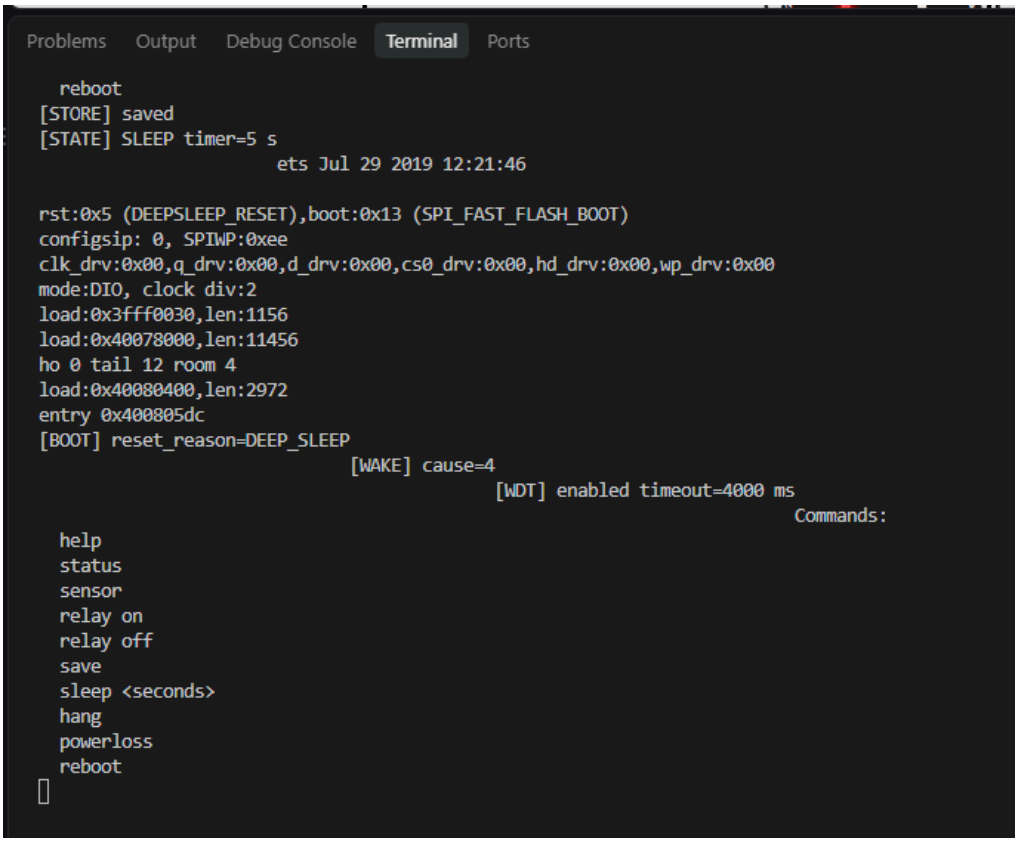
Field	Value
MCU	ESP32 DevKit V1
Sensor	DHT22 GPIO15
Button	GPIO4
LED	GPIO2
Relay	GPIO5
Serial	115200
Build	BUGGY

Test Matrix

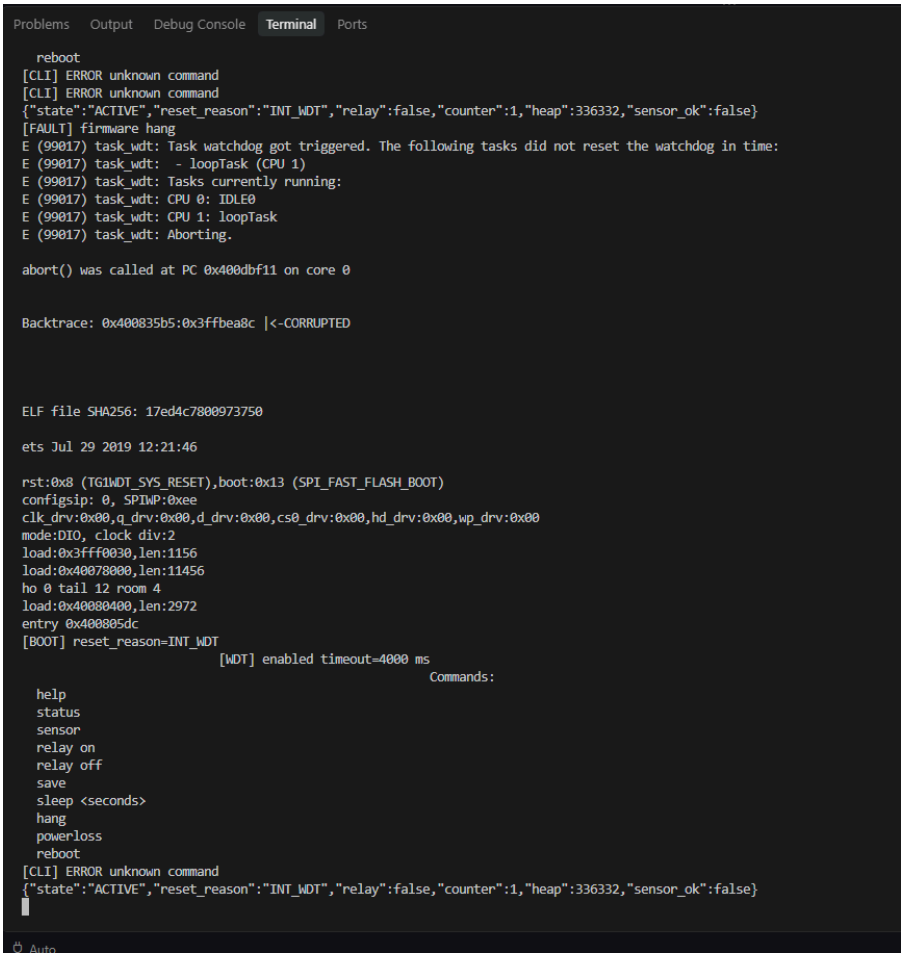
Scenario	Risk	Precondition	Action	Expected	Actual	Evidence	P/F
Sleep 5 (Sleep timing)	Timing / wake / persistence (Wrong state timing)	Device ACTIVE	`sleep 5`	Wake in 4.25–5.75 s	Wake in less than 1s	Timestamped log + video https://drive.google.com/file/d/1sDcnInEVqLWMa99T-0tNufTBgGRBEwIy/view?usp=sharing	F bug_1
WDT hang (Device unavailable)	Reset time / reason / recovery (Firmware hang)	Device ACTIVE	`hang`	WDT reset ≤5 s	WDT reset in ~30s	Reset log https://drive.google.com/file/d/1NZsC2oouEr1XDFYsLc21fo2NwUm5exFT/view?usp=sharing	F bug_2 (WITHDRAWN, see note)
Relay + WDT (Unsafe actuator state)	Safe actuator state (Unsafe recovery)	Relay ON	`hang`	After WDT relay OFF	After WDT relay ON	Screen + status	F bug_3
Interrupted save	Atomicity / integrity (Corrupted persistent state)	Known state	`powerloss	Old or new valid state only	Counter is changed, relay is not changed	Before/after status	?
Sensor failure (Sensor unavailable)	Explicit invalid-data handling (Invalid sensor data)	Device ACTIVE	Disconnect DHT	Explicit sensor error	[SENSOR] temp=0.0C humidity=0.0%	Serial log	F bug_4
Button (Button bounce)	Single logical event (Duplicate event)	Device ACTIVE	Press button 20 times	Exactly 20 events	Object non testable	Counter + video	N/A
Repeated recovery (Recovery stability)	No boot loop / stable (Boot loop)	Device ACTIVE	hang/recovery ×10	Returns ACTIVE each time	"State": "ACTIVE", "reset_reason":"INT_WDT" After each of 10 cycles	Serial log	P

Screenshots

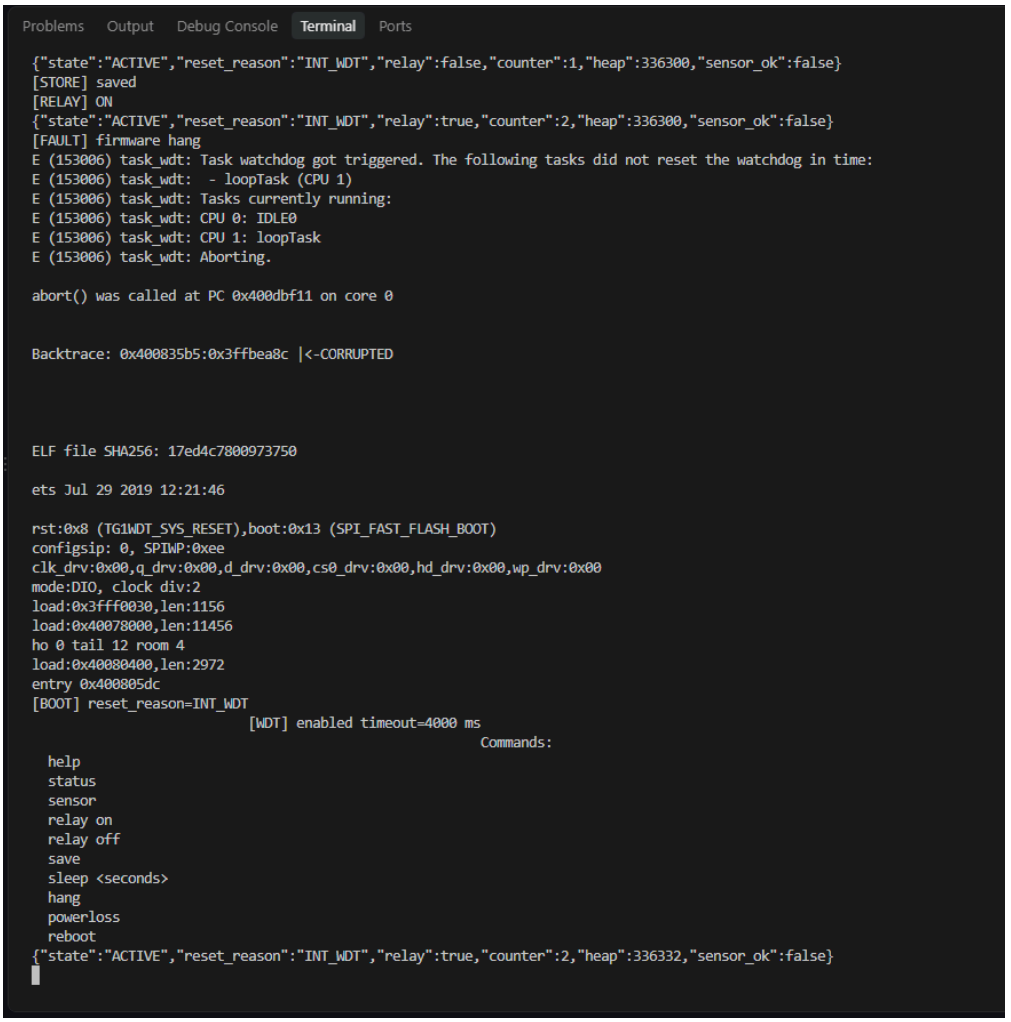
Sleep timing



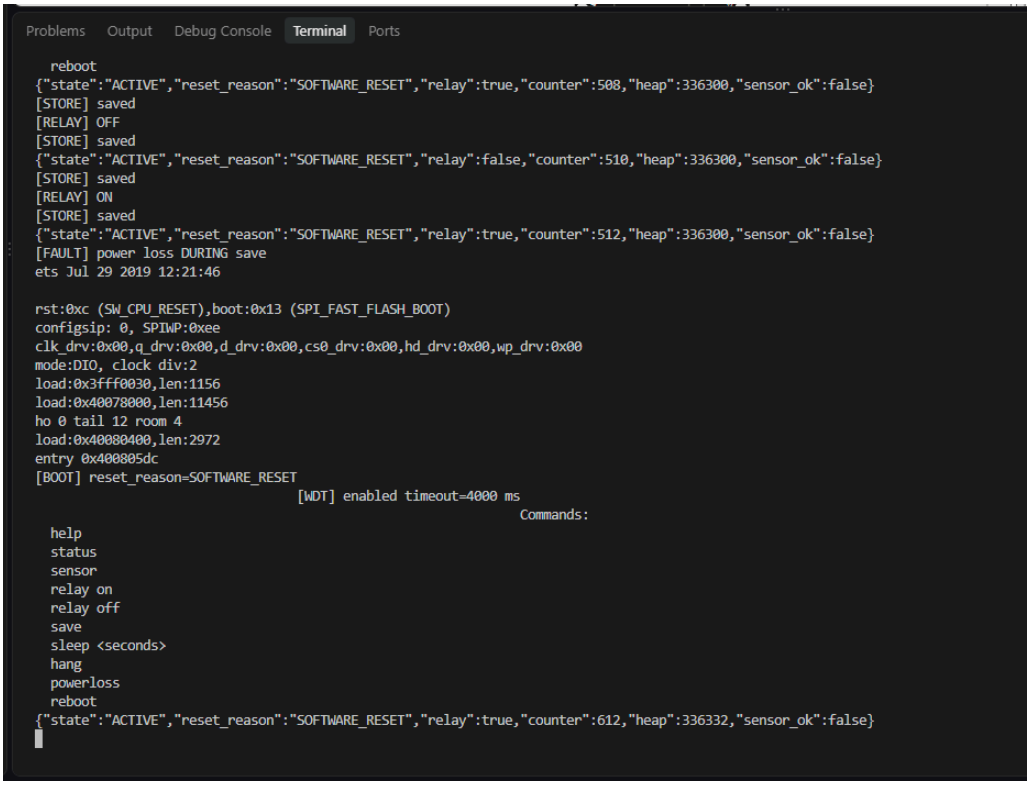
Device unavailable



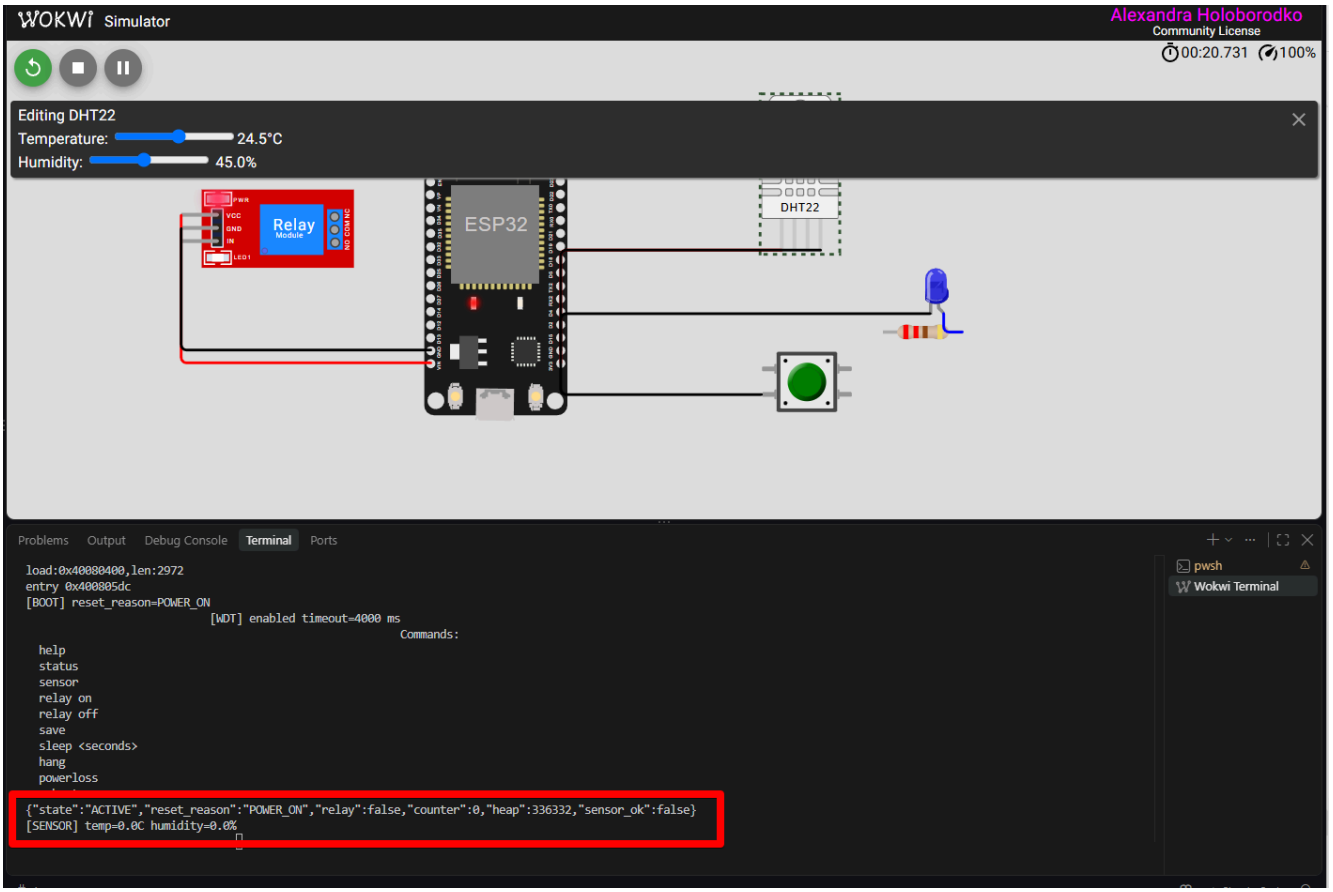
Relay + WDT



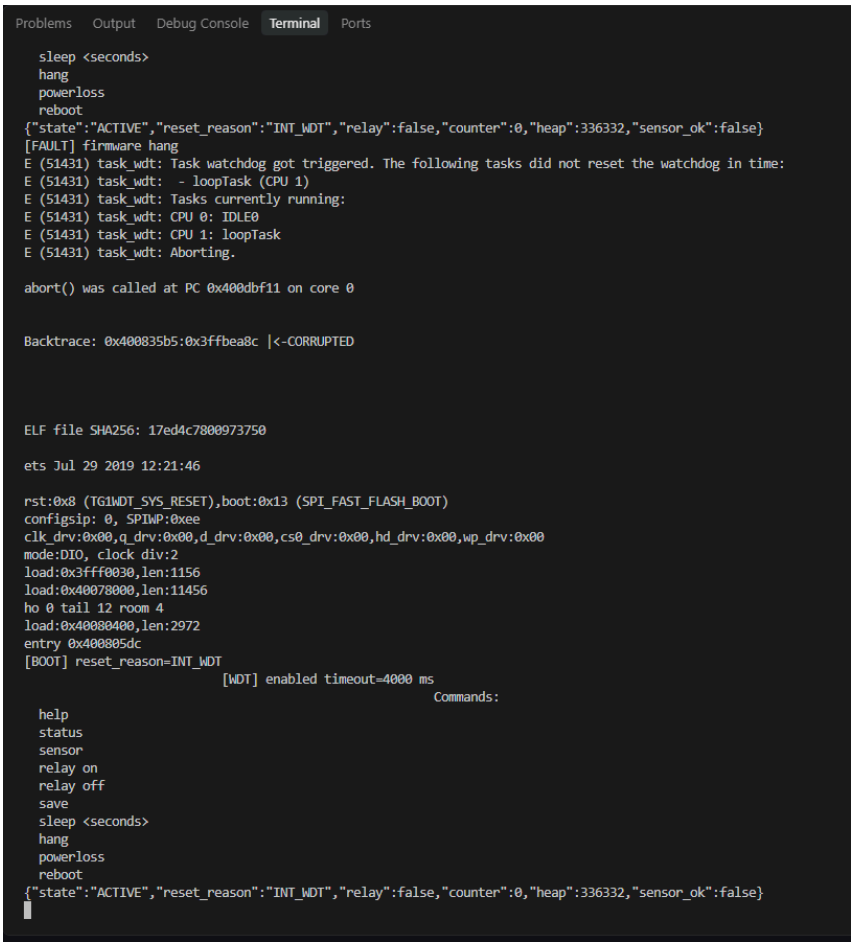
Interrupted save



Sensor unavailable



Recovery stability



Bug Reports

Bug_1

Title: System wakes in less than 1s after entering “sleep 5” command

Requirement: The `sleep <N>` command puts the device into deep sleep for approximately N seconds.

Risk: Wrong state timing

Preconditions: Device ACTIVE

Steps:

1. Enter “sleep 5” command
2. Check time when system wakes

Expected: System wakes in 4.25–5.75 s

Actual: System wakes in less than 1s

Timing / Reset Reason: deep sleep

Evidence: serial log, https://drive.google.com/file/d/1sDcnInEVqLWMa99T-0tNufTBgGRBEwIy/view?usp=sharing

Severity: major

Reproducibility: 5/5

Bug_2 — WITHDRAWN (see note at the top of the report)

Title: WDT reset is in ~30s instead of ≤5 s

Requirement: The WDT reboots the device within a maximum of approximately 5 seconds

Risk: Firmware hang

Preconditions: Device ACTIVE

Steps:

1. Enter “hang” command
2. Check log

Expected: WDT reset ≤5 s

Actual: WDT reset in ~30s

Timing / Reset Reason: INT_WDT

Evidence: serial log, https://drive.google.com/file/d/1NZsC2oouEr1XDFYsLc21fo2NwUm5exFT/view?usp=sharing

Severity: major

Reproducibility: 5/5

Bug_3

Title: Relay is ON after entering “hang” command

Requirement: The actuator must be OFF after an emergency reset

Risk: Unsafe recovery

Preconditions: Device ACTIVE, Relay ON

Steps:

1. Enter “hang” command
2. Enter “status” command
3. Check log

Expected: After WDT relay OFF

Actual: After WDT relay ON

Timing / Reset Reason: INT_WDT

Evidence: serial log

Severity: critical

Reproducibility: 5/5

Bug_4

Title: “Sensor” command returns data when sensor is disconnected

Requirement:

If the DHT22 is unavailable or returns incorrect data:

- the firmware must not treat this as a valid measurement;

- there must be an explicit error in the logs;

Risk: Invalid sensor data

Preconditions: Device ACTIVE

Steps:

1. Disconnect DHT

2. Enter “sensor” command

3. Check log

Expected: Explicit sensor error

Actual: [SENSOR] temp=0.0C humidity=0.0%

Timing / Reset Reason: POWER_ON

Evidence: serial log

Severity: critical

Reproducibility: 5/5

Release Decision

Decision	**NO-GO**
Blocking risks	Wrong state timing (bug_1) Firmware hang (bug_2 (WITHDRAWN, see note)) Unsafe recovery (bug_3) Invalid sensor data (bug_4)
Tests required on real hardware	Deep sleep power consumption Button debounce Power-loss recovery Sensor error handling Wake-up reliability